(36)

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I V THANISHA/196526016 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: TVJ .

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	2	2	2
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	1	1	1

Total Marks 24 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

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COURSE: COMPUTER NETWORKS

COURSE CODE: C6A0709

DEPT: CSC (CYBER)

29

30

ERROR CONTROL

OVERALL MECHANISM THAT ENSURES DATA IS TRANSMITTED ACCURATELY AND RELIABLY ACROSS THE NETWORK

DATA LINK LAYER

HANDLES NODE TO NODE COMMUNICATION AND PROVIDE ERROR CONTROL AT THE LINK LEVEL

KEY FUNCTIONS

* FRAMES DATA
* USES ERROR DETECTION TECHNIQUES
* DETECTS ERRORS

ERROR DETECTION

IDENTIFIES IF ERRORS HAVE OCCURRED IN THE RECEIVED DATA

HOW IT WORKS

* CALCULATES A CHECKSUM.
* COMPARES WITH RECEIVED DATA
* FLAG ERRORS

TRANSPORT LAYER

PROVIDES END-TO-END COMMUNICATION AND ENSURE DATA IS DELIVERED CORRECTLY BETWEEN APPLICATIONS

KEY FUNCTIONS

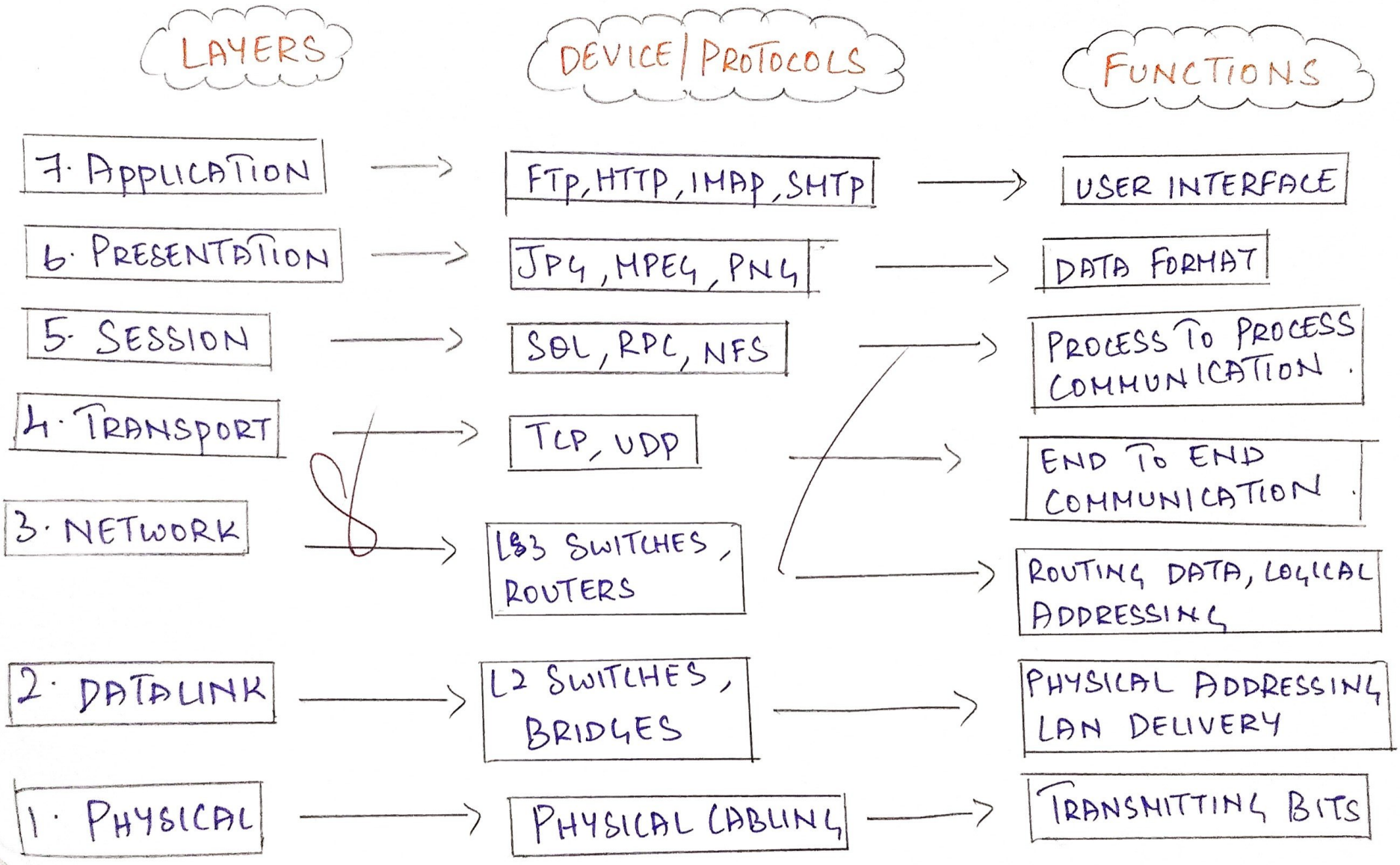
* USES ERROR CONTROL MECHANISMS.
* MANAGES LOST/CORRUPTED DATA
* ENSURES PROPER DATA FLOW AND ORDERING.

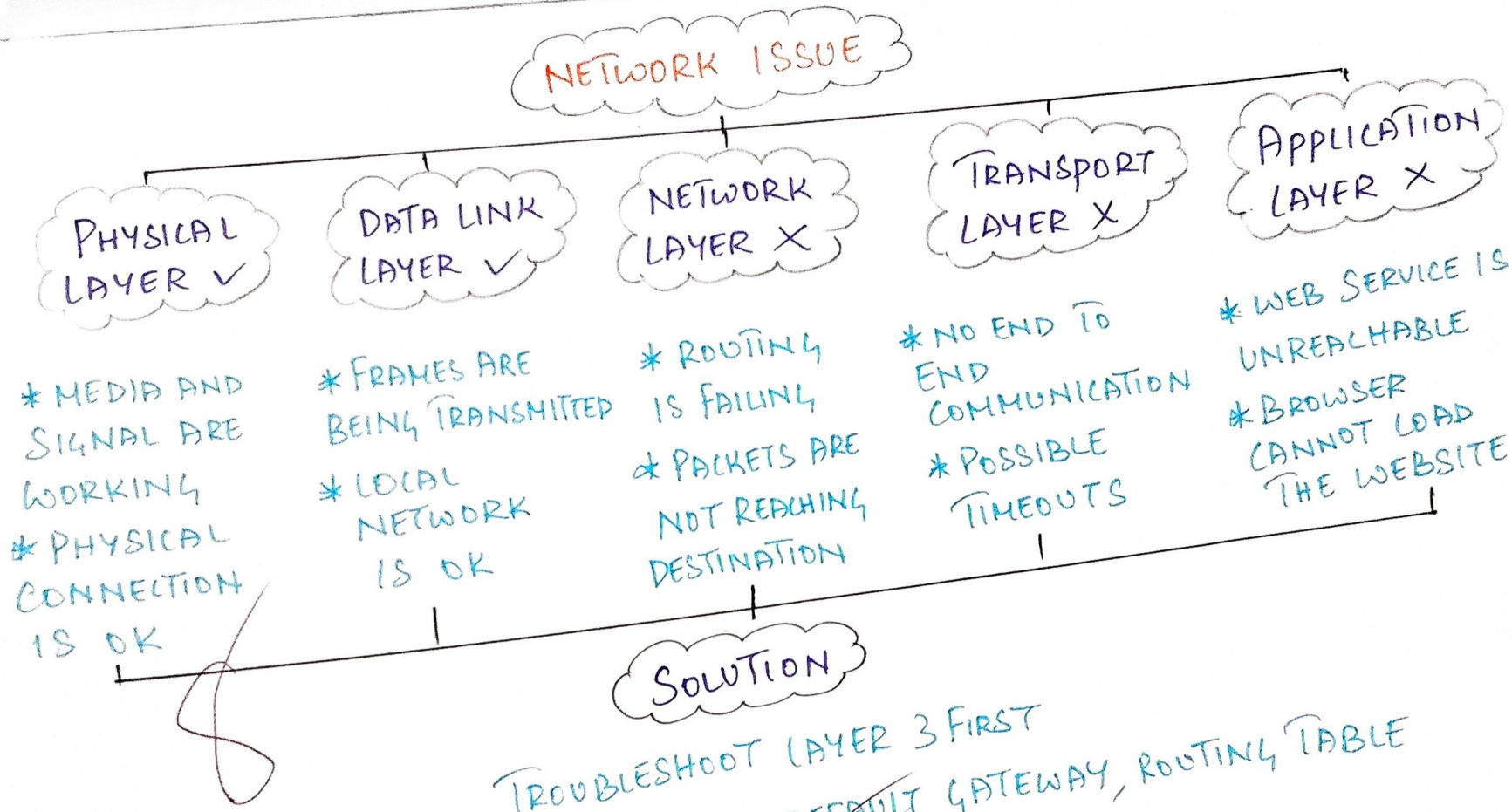
ERROR CORRECTION

FIXES ERRORS IN THE RECEIVED DATA SO THE INFORMATION IS ACCURATE AND COMPLETE.

HOW IT WORKS

* USES TECHNIQUES LIKE RETRANSMISSION OR FORWORD ERROR CORRECTION
* REQUEST MISSING DATA TO BE RESENT OR CORRECT IT DIRECTLY.


LAYERS
DEVICE/PROTOCOLS
FUNCTIONS
7. APPLICATION
FTP, HTTP, IMAP, SMTP
USER INTERFACE
6. PRESENTATION
JPG, MPEG, PNG
DATA FORMAT
5. SESSION
SQL, RPC, NFS
PROCESS TO PROCESS COMMUNICATION
4. TRANSPORT
TCP, UDP
END TO END COMMUNICATION
3. NETWORK
L3 SWITCHES, ROUTERS
ROUTING DATA, LOGICAL ADDRESSING
2. DATALINK
L2 SWITCHES, BRIDGES
PHYSICAL ADDRESSING LAN DELIVERY
1. PHYSICAL
PHYSICAL CABLING
TRANSMITTING BITS



TROUBLESHOOT LAYER 3 FIRST

CHECK IP CONFIGURATION, DEFAULT GATEWAY, ROUTING TABLE AND CONNECTIVITY TO THE DESTINATION